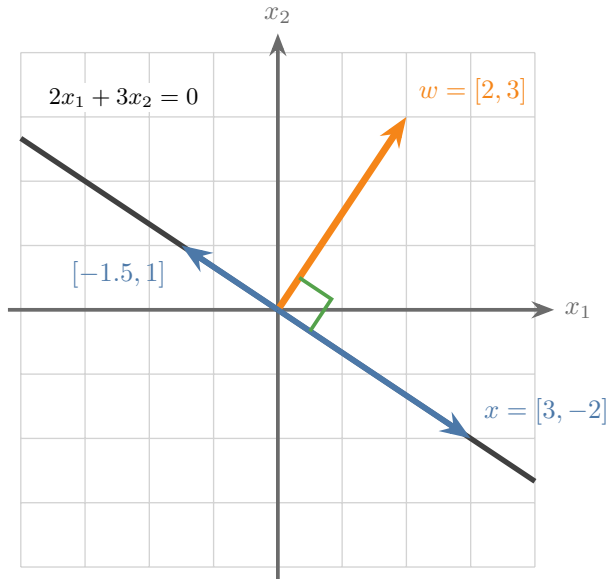
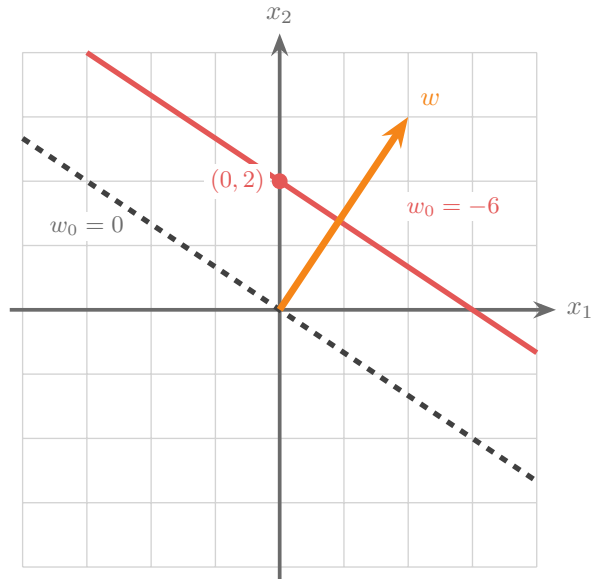


Line through the origin: $w^\top x = 0$



Adding w_0 moves the line



$w \cdot x = 6 - 6 = 0$: w is at 90° to every x on the line

$2x_1 + 3x_2 - 6 = 0$: same w , parallel line, intercept $-w_0/w_2 = 2$